9.5 Making Change

WORKSHEET

The question

STEP I 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How many ways are there to make n pence using only 2p and 1p coins? Why does the answer depend only on how many 2p coins you use?

STAGE 1 Ten pence

Covers: making 10 pence

Ask yourself Sort by the largest coins. Either you use a 10p coin, or you don't. If you don't, how many 5p coins can you use? Then count the ways to make the rest with 2p and 1p coins.

STAGE 2 Building up layer by layer

Covers: making 20 pence

Pause and try Let $b(n)$ be the number of ways to make n pence from 5p, 2p and 1p coins. Choose the number of 5p coins, and write $b(n)$ as a sum of values of the 2p-and-1p count. Make a table for $n = 0, 5, 10, 15, 20$.

Ask yourself Now bring in the 10p coins in the same way, using your values of b.

STAGE 3 Twenty pence

Covers: making 20 pence

? **Ask yourself** Last of all: either you use the 20p coin or you don't. Add up.

What's really going on

Coins as geometric series

Here is the idea that turns this into algebra. The number of 5p coins used can be $0, 1, 2, \ldots$, contributing $0, 5, 10, \ldots$ pence. Record this as the series

$$1 + x^5 + x^{10} + x^{15} + \cdots = \frac{1}{1 - x^5}.$$

Do the same for each coin and multiply:

$$\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})(1-x^{20})} = \sum_n w(n)x^n.$$

When the brackets are multiplied out, each term x^n arises from choosing one term from each bracket, that is, a number of each coin, with total value n. So $w(n)$, the coefficient of x^n, is exactly the number of ways to make n pence. The table in Stage 2 was simply multiplying out this product one bracket at a time. This is the generating function idea of Section 9.4, and multiplication is where it shows its power: products of generating functions count combinations of choices.

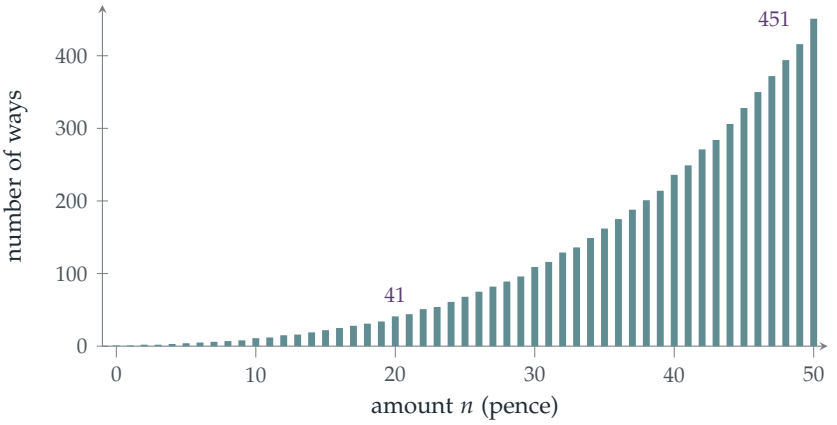


Figure 9.5.1 The number of ways to make n pence from 1p, 2p, 5p, 10p, 20p and 50p coins. The jumps at multiples of 5 and 10 show where a new coin first becomes usable. With six coins the count grows roughly like n^5: in general, like a polynomial of degree one less than the number of coins.

Partitions

What if every whole number is allowed as a coin? Then the number of ways to write n as a sum of positive integers, ignoring order, is the *partition number* $p(n)$. For example $p(4) = 5$, since $4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$. Euler saw in 1748 that

$$\sum_{n \geq 0} p(n) x^n = \frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)\cdots},$$

and used it to discover remarkable identities. One of them: the number of ways to write n as a sum of *distinct* parts equals the number of ways to write it as a sum of *odd* parts. (The proof is a single line of algebra with generating functions: try it in the exercises.)

Hardy, Ramanujan and $p(200)$

The partition numbers grow fast: $p(10) = 42$, $p(100) = 190\,569\,292$. In 1918 G. H. Hardy and Srinivasa Ramanujan found an astonishing formula for $p(n)$, an infinite series whose first term is

$$p(n) \approx \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{2n/3}}.$$

To test it, Percy MacMahon computed $p(200) = 3\,972\,999\,029\,388$ by hand, using Euler's identities. The first few terms of Hardy and Ramanujan's series gave the right answer to within a fraction of one. Ramanujan also discovered that $p(5n+4)$ is always divisible by 5, $p(7n+5)$ by 7, and $p(11n+6)$ by 11. Explaining why there are no more such patterns took until the twenty-first century.

Try this next

1 **Two kinds of six.** Try STEP II 2012, Question 1 (step.maths.org/questions/12-s2-q1). It counts with products of geometric series. At the end, describe a coin-counting problem that the coefficients answer.

2 **Distinct and odd.** Show that $(1+x)(1+x^2)(1+x^3)\cdots$ is the generating function for partitions into distinct parts. Use $1+x^k = \frac{1-x^{2k}}{1-x^k}$ to show that it equals $\frac{1}{(1-x)(1-x^3)(1-x^5)\cdots}$, and deduce Euler's theorem. Check it for $n = 6$.

3 **A formula.** Show that the number of ways to make n pence using only 1p, 2p and 5p coins is the nearest integer to $\frac{(n+4)^2}{20}$. (Check it first for small n.)

4 **Order matters.** In how many ways can you make n pence by putting 1p and 2p coins into a slot one at a time, if the order matters? (You have met this sequence before.)

Further reading

- **George Andrews & Kimmo Eriksson, *Integer Partitions*** (Cambridge University Press). A short, readable introduction for undergraduates.
- **George Pólya, "On picture-writing"**, *American Mathematical Monthly* 63 (1956). Pólya's classic explanation of generating functions using the problem of changing a dollar (there are 292 ways).
- **Robert Kanigel, *The Man Who Knew Infinity*** (Abacus). The life of Ramanujan, and his partnership with Hardy.

The full guided solution is in Section 9.5:
https://beyondstep.keithmcnulty.org/ch09/making-change.html